

CHAPTER 1. Introduction

This report focuses on Exploratory Data Analysis (EDA) and its feature engineering in the Student Habits dataset. Through the discussion, it is tried to explore which factors are related to students' test scores, how the combination of these factors affects the scores, and what patterns students who get good grades usually have.

1.1 Dataset Introduction

The student habits dataset is referenced from Kaggle [1]. This is a simulated dataset exploring how lifestyle habits affect academic performance in students. With 1,000 student records and features including study hours, sleep patterns, social media usage, diet quality, mental health, and more - against their final exam score.

CHAPTER 2. EDA

The main questions to answer in the EDA section are:

1. Are there missing/duplicate values in the dataset?
2. What are the overall metrics (mean, maximum, minimum, etc.) for each habit and final exam score? How are they distributed? (For example, is it normal, skewed, or something else?) Are there any outliers?
3. Is there a correlation between each habit and the final exam score?
4. Are there other habits that are correlated with each other? For example, is the amount of time spent studying correlated with the amount of time spent sleeping?
5. Are there specific groups of students or patterns that are associated with high or low academic performance?

2.1 Data Overview and Distribution Analysis

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 16 columns):
 #   Column                                  Non-Null Count  Dtype
---  -
 0   student_id                             1000 non-null   object
 1   age                                     1000 non-null   int64
 2   gender                                 1000 non-null   object
 3   study_hours_per_day                    1000 non-null   float64
 4   social_media_hours                     1000 non-null   float64
 5   netflix_hours                          1000 non-null   float64
 6   part_time_job                           1000 non-null   object
 7   attendance_percentage                  1000 non-null   float64
 8   sleep_hours                           1000 non-null   float64
 9   diet_quality                           1000 non-null   object
10   exercise_frequency                     1000 non-null   int64
11   parental_education_level                909 non-null    object
12   internet_quality                       1000 non-null   object
13   mental_health_rating                   1000 non-null   int64
14   extracurricular_participation           1000 non-null   object
15   exam_score                             1000 non-null   float64
dtypes: float64(6), int64(3), object(7)
memory usage: 125.1+ KB
None
```

Figure 1 Information of Dataset

	age	study_hours_per_day	social_media_hours	netflix_hours	attendance_percentage	sleep_hours	exercise_frequency	mental_health_rating	exam_score
count	1000.0000	1000.00000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	20.4980	3.55010	2.505500	1.819700	84.131700	6.470100	3.042000	5.438000	69.601500
std	2.3081	1.46889	1.172422	1.075118	9.399246	1.226377	2.025423	2.847501	16.888564
min	17.0000	0.00000	0.000000	0.000000	56.000000	3.200000	0.000000	1.000000	18.400000
25%	18.7500	2.60000	1.700000	1.000000	78.000000	5.600000	1.000000	3.000000	58.475000
50%	20.0000	3.50000	2.500000	1.800000	84.400000	6.500000	3.000000	5.000000	70.500000
75%	23.0000	4.50000	3.300000	2.525000	91.025000	7.300000	5.000000	8.000000	81.325000
max	24.0000	8.30000	7.200000	5.400000	100.000000	10.000000	6.000000	10.000000	100.000000

Figure 2 Describe of Dataset

First, let's look at the information of the dataset. As shown in Figure 1, about 9.1% of the data in the `parental\_education\_level` column is missing, and there are no duplicate records in the dataset.

From the descriptive statistics (Figure 2), it can be seen that the average age is 20.5 years old, the standard deviation is 2.3, and the range is between 17 and 24 years old.

The average study time (study_hours_per_day) is 3.65 hours, and the standard deviation is 1.47, indicating that there are significant differences in students' study habits. The average social media usage time (social_media_hours) is 2.61 hours, while the average Netflix viewing time is 1.82 hours, indicating that students are more diverse in their time allocation for entertainment activities.

count	
sleep_hours	
6 to 8	513
less than 6	368
more or equal to 8	119

Figure 3 Counting of Sleeping Hours

The average sleep time is 6.47 hours. According to the sleep time counting (Figure 4), most students sleep for 6 to 8 hours. If 8 hours is used as the standard, this may indicate that most students do not get enough sleep, but if 6 hours is used as the minimum standard, it is in line with the sleep pattern of students.

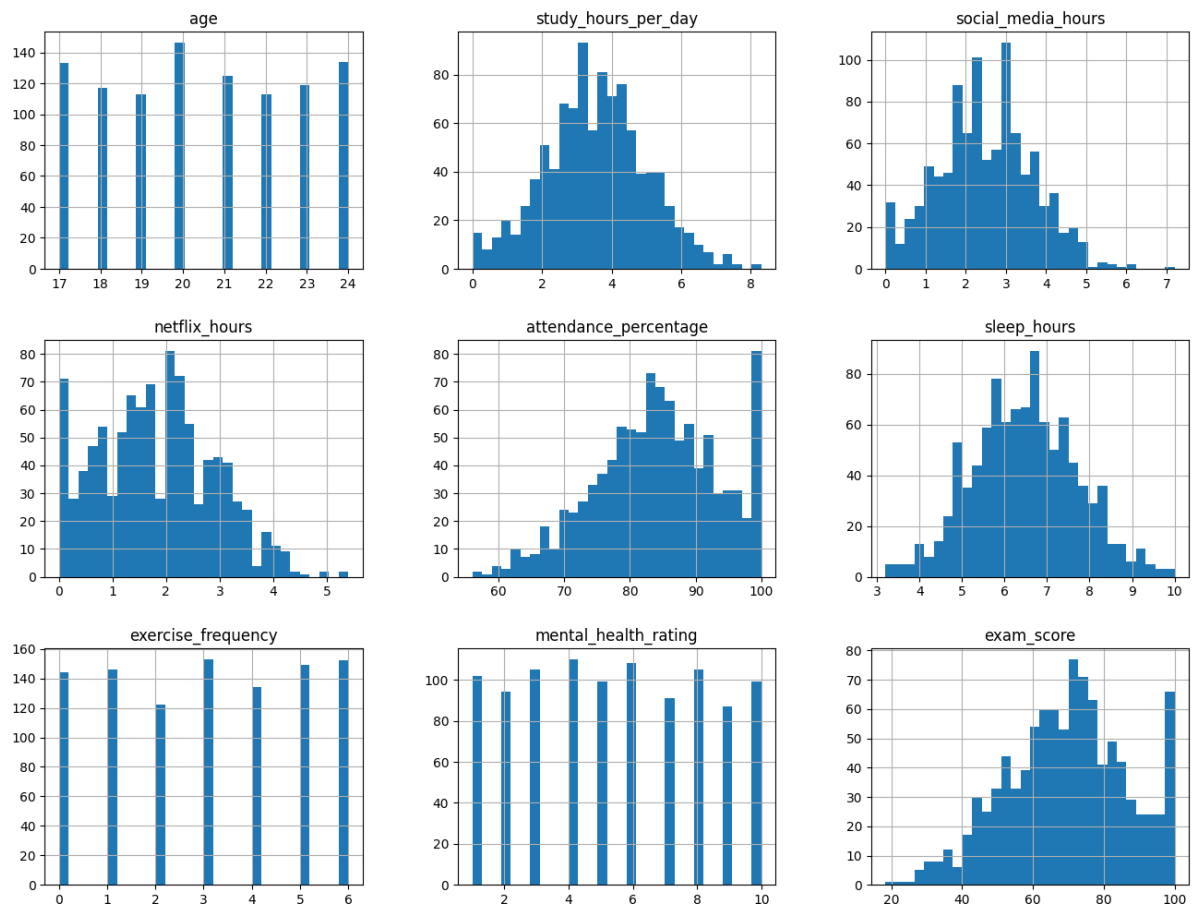


Figure 4 Distribution of Numerical Variables

```

age: 0.0084371397444062
study_hours_per_day: 0.05425310140811603
social_media_hours: 0.11980520318308562
netflix_hours: 0.23715439564540441
attendance_percentage: -0.23781042684597256
sleep_hours: 0.09148397227093791
exercise_frequency: -0.03192297249384373
mental_health_rating: 0.037810699888038285
exam_score: -0.15635065547132798

```

Figure 5 Skewness of Distribution of Numerical Variables

From the distribution plots (Figure 5) and skewness (Figure 6), we can see that the numerical variables of the dataset show a relatively normal distribution (skewness between -0.5 and 0.5).

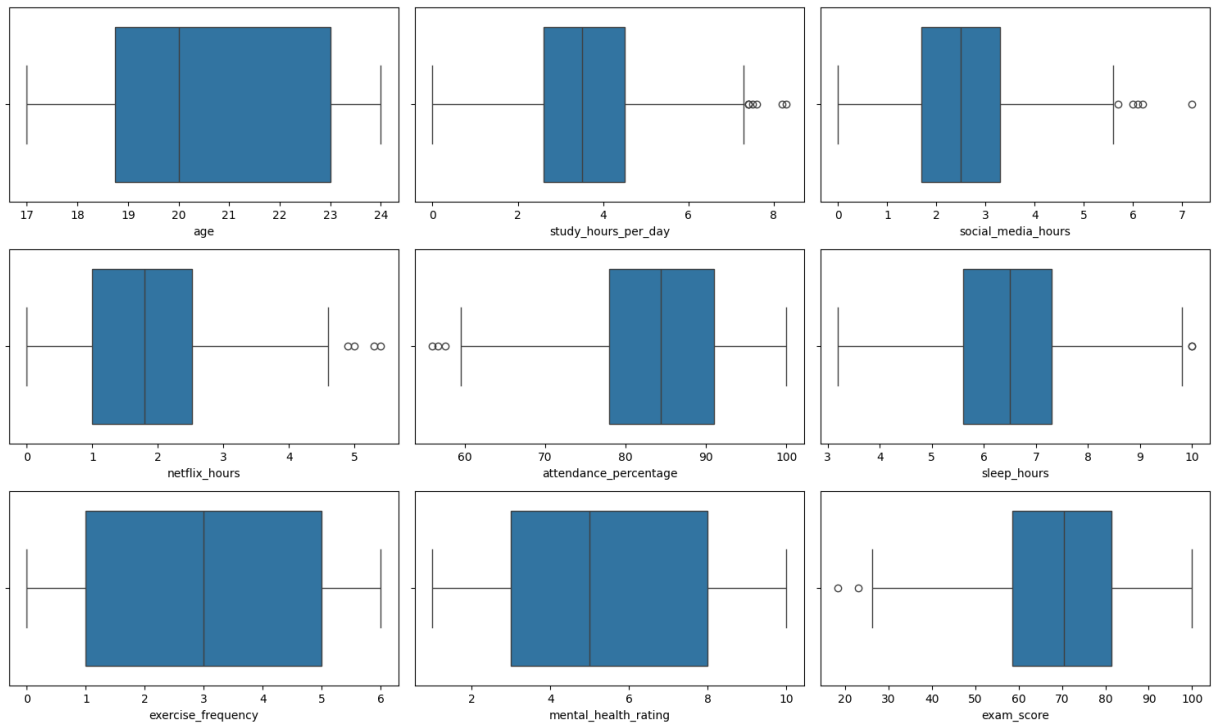


Figure 6 Boxplot of Numerical Variables

From the box plot analysis (Figure 7), it can be seen that most numerical variables have some outliers. In particular, there are obvious upper-end outliers in terms of study time, social media usage time, and Netflix viewing time, indicating that a few students have invested extreme time in these activities.

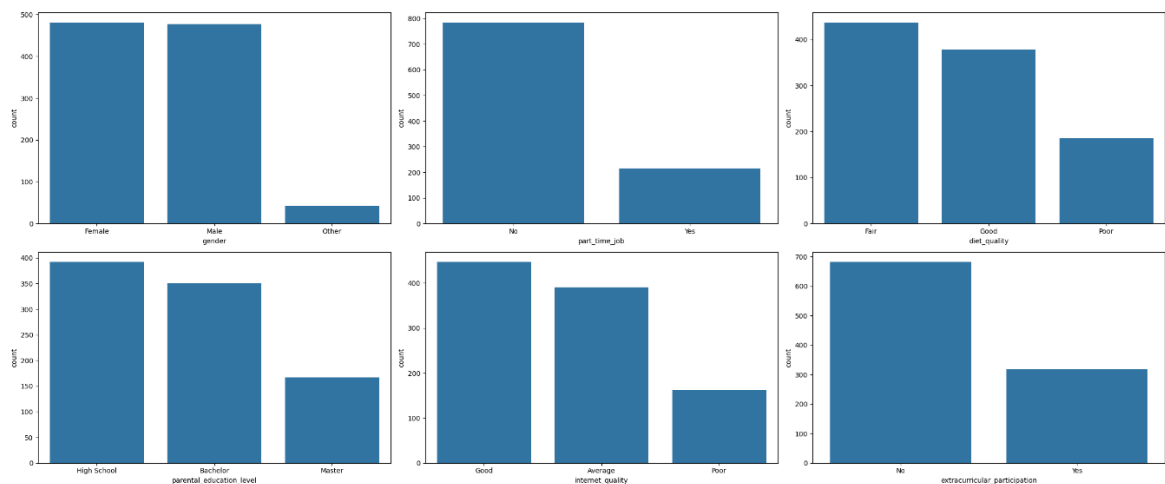


Figure 7 Counting of Category Variables

The bar plot of categorical variables shows varying degrees of uneven distribution, which may have an important impact on academic performance. It will be further discussed in the hypothesis testing section.

2.2 Correlation Analysis Between Habits and Exam Scores

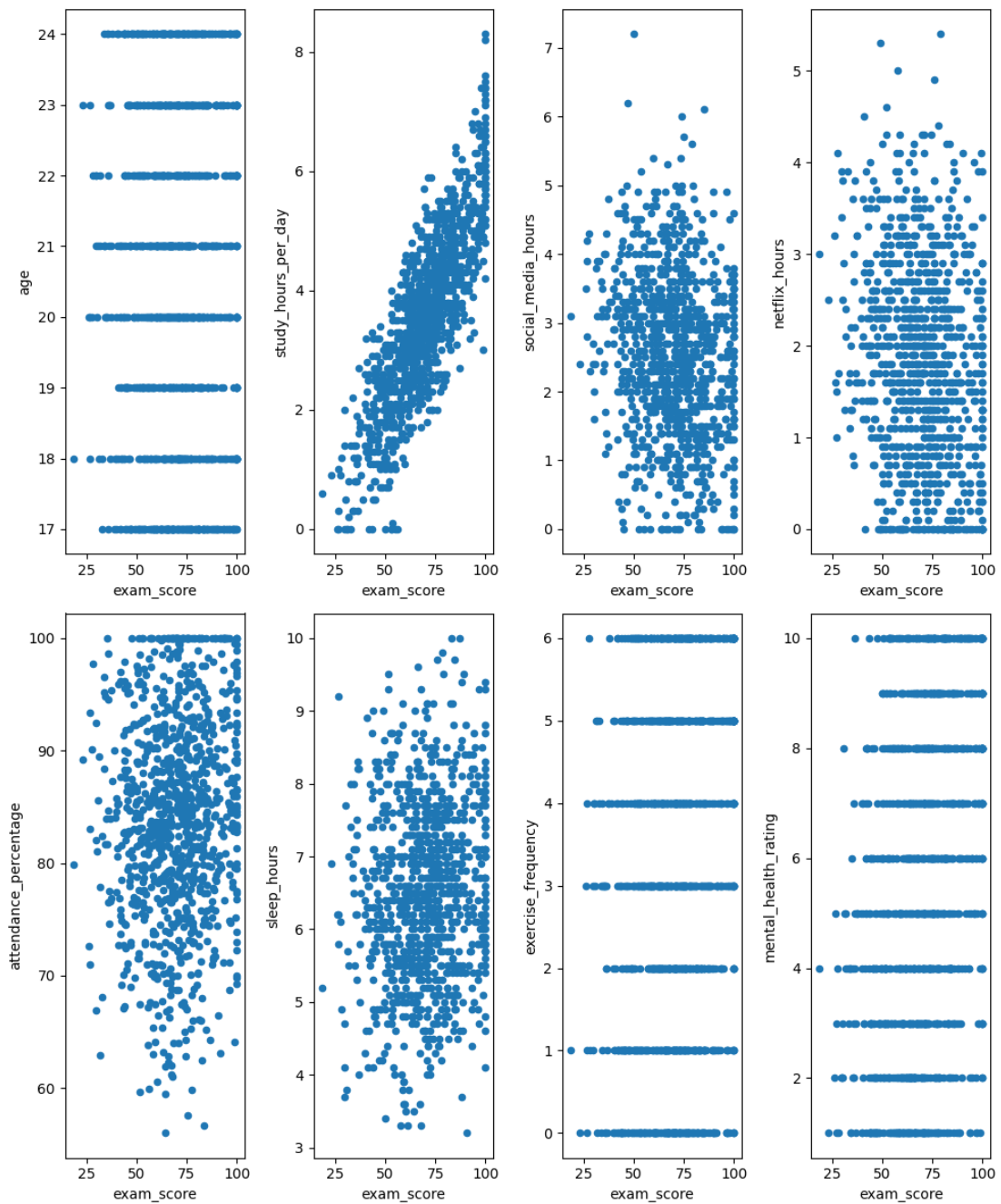


Figure 8 Scatter Plot of Numerical Variables

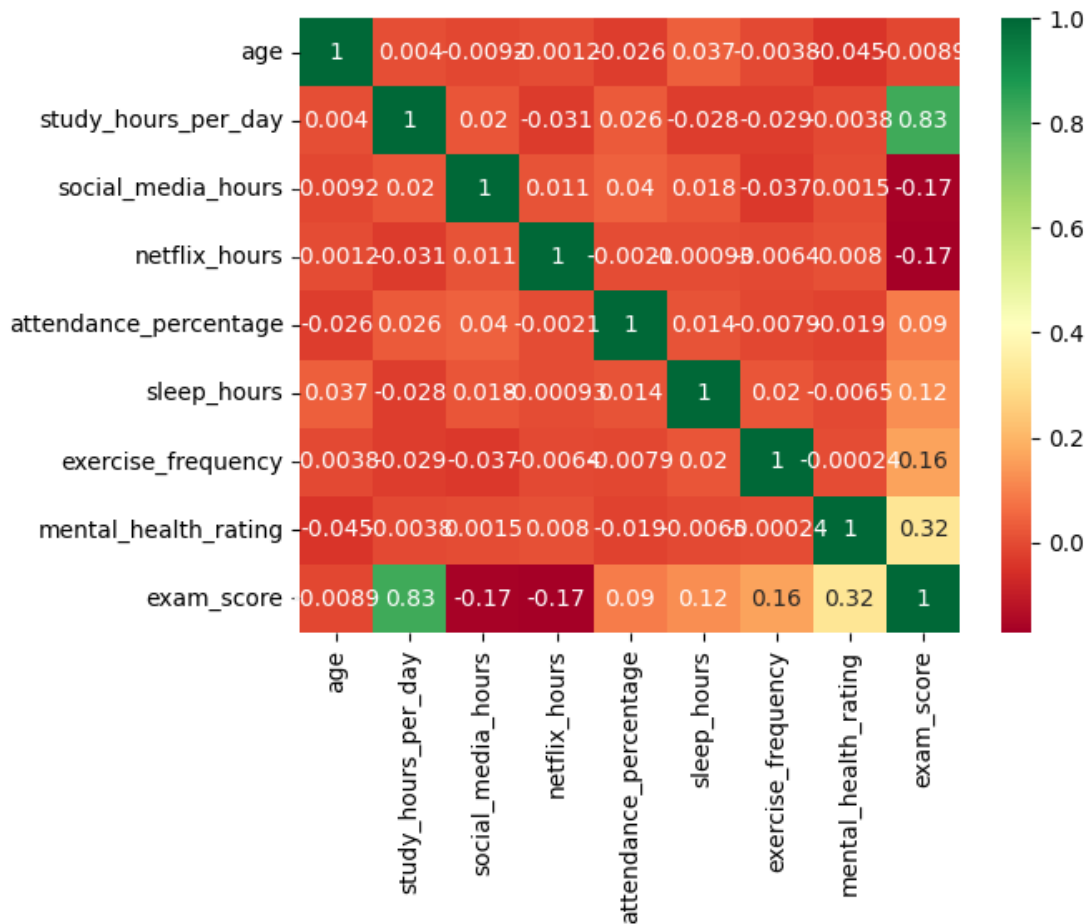


Figure 9 Heatmap of Numerical Variables

It is clear to show from the correlation heat map, there is the strongest positive correlation between study hours per day and exam scores, with a correlation coefficient of 0.83, which is the most significant correlation among all variables. This finding is consistent with the basic intuition that spending more time in study generally leads to better academic performance.

Mental health rating also shows a moderate positive correlation with exam scores ($r=0.32$), which may indicate that a good mental state has a positive impact on academic performance. Exercise frequency shows a weak positive correlation with the exam scores ($r=0.16$). Although the correlation is not strong, it suggests that moderate physical exercise may help improve academic performance.

Both social media usage time (social_media_hours) and Netflix viewing time (netflix_hours) show a negative correlation with exam scores ($r=-0.17$), indicating that entertainment activities may distract students and affect learning outcomes. The correlation between sleep time (sleep_hours) and test scores is weak ($r=0.12$), which may be because sleep quality is more important than sleep duration.

The correlation between attendance (attendance_percentage) and test scores is 0.09, which is weak but still positive, this indicates that class participation may have certain positive effect on academic performance. There is almost no correlation between age

and exam scores, which indicates that within this age range, age differences have little effect on academic performance.

Although most habits are not highly correlated with the exam scores, the impact of these habits can still be further discussed in the hypothesis testing section.

2.3 Correlation Analysis Between Different Habits

Through the analysis of the correlations heatmap between the habit variables, we found some interesting patterns. First, the relationship between study time and other variables is relatively independent, which indicates that study habits may be a relatively independent behaviour pattern and are not directly affected by other life habits.

There is no significant correlation between social media use time and Netflix viewing time ($r=0.02$), which indicates that students may have different preferences between different types of entertainment activities.

2.4 Student Group Pattern Analysis and Hypothesis Testing

2.4.1 Hypothesis Testing of Category Variables

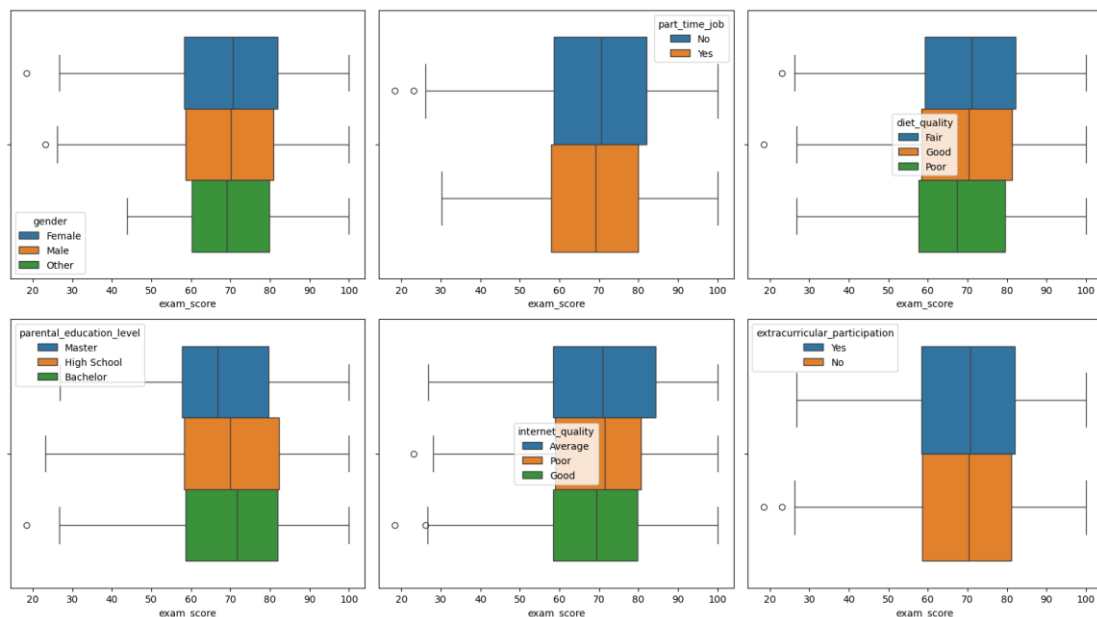


Figure 10 Boxplot of Exam Score under Category Variables

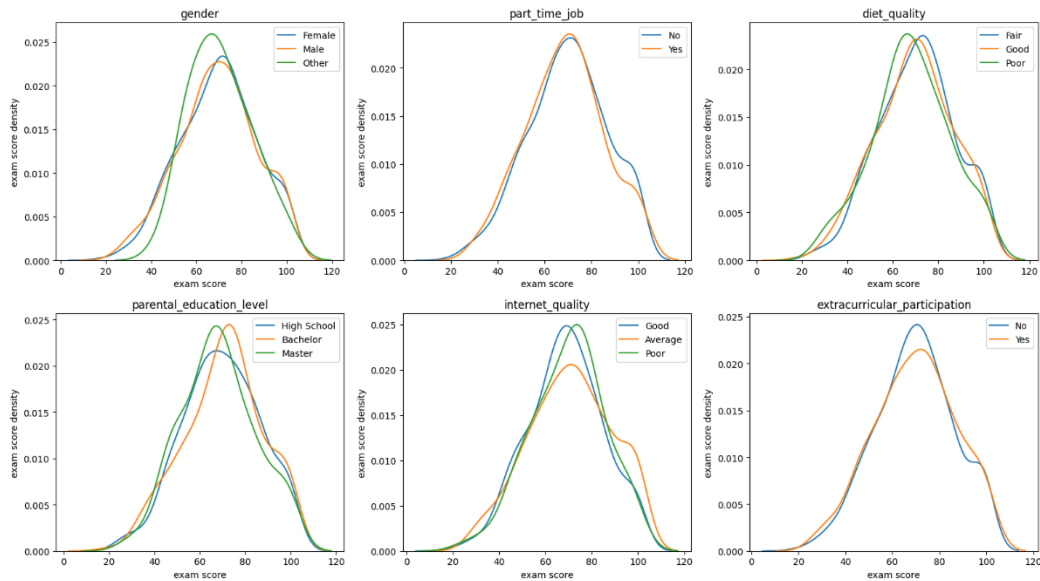


Figure 11 Distribution Plot of Exam Scores under Category Variables

With the help of the correlation heat map and the box plot and distribution plot of the categorical variables (Figures 10 and 11), we can conduct some hypothesis tests to check whether the categorical variables have a significant impact on the test scores. The significance level α is set to 0.05, and for the `parental_education_level` column with null values, we delete the null values before conducting the hypothesis test.

For each categorical variable, the null hypothesis in the hypothesis test is to assume that there is no significant difference in the test scores under different categories of the categorical variable, and the alternative hypothesis is that there is a significant difference. Taking `gender` as an example, the null hypothesis is to assume that there is no difference in the test scores of males, females, and others ($\mu_{\text{male}} = \mu_{\text{female}} = \mu_{\text{other}}$), and the alternative hypothesis is to assume that there is a difference between them ($\mu_{\text{male}} \neq \mu_{\text{female}} \neq \mu_{\text{other}}$). In addition, for each categorical variable, if its number of categories is 2, a two-sample t-test is performed to compare the differences in test scores between different groups; for variables with more than two categories, a one-way analysis of variance (ANOVA) is used.

```
skewness of exam score under gender-Female: -0.16051025436666994
skewness of exam score under gender-Male: -0.16719118207224112
skewness of exam score under gender-Other: 0.3467256718743277
skewness of exam score under part_time_job-No: -0.1905640177007152
skewness of exam score under part_time_job-Yes: -0.03071885608246065
skewness of exam score under diet_quality-Fair: -0.14478508719808597
skewness of exam score under diet_quality-Good: -0.1917098294774159
skewness of exam score under diet_quality-Poor: -0.0981298843160782
skewness of exam score under parental_education_level-High School: -0.13898002062787018
skewness of exam score under parental_education_level-Bachelor: -0.25807126242239753
skewness of exam score under parental_education_level-Master: 0.06961953706500562
skewness of exam score under internet_quality-Good: -0.13027549885669606
skewness of exam score under internet_quality-Average: -0.17767763046861448
skewness of exam score under internet_quality-Poor: -0.27883406354516777
skewness of exam score under extracurricular_participation-No: -0.1517313801228753
skewness of exam score under extracurricular_participation-Yes: -0.16607634768463425
Result: At the  $\alpha=0.05$  level, gender - 0.8673937379445218 has no significant effect on exam scores
Result: At the  $\alpha=0.05$  level, part_time_job - 0.8673937379445218 has no significant effect on exam scores
Result: At the  $\alpha=0.05$  level, diet_quality - 0.28235232253240805 has no significant effect on exam scores
Result: At the  $\alpha=0.05$  level, parental_education_level - 0.3901114487005922 has no significant effect on exam scores
Result: At the  $\alpha=0.05$  level, internet_quality - 0.2320346510834057 has no significant effect on exam scores
Result: At the  $\alpha=0.05$  level, extracurricular_participation - 0.2320346510834057 has no significant effect on exam scores
```

Figure 12 Skewness and Results of Hypothesis Testing - 1

According to the results of hypothesis testing, we can see that all the category variables do not have significant effect on exam scores.

2.4.2 Pattern Analysis and Hypothesis Testing of Numerical Variables

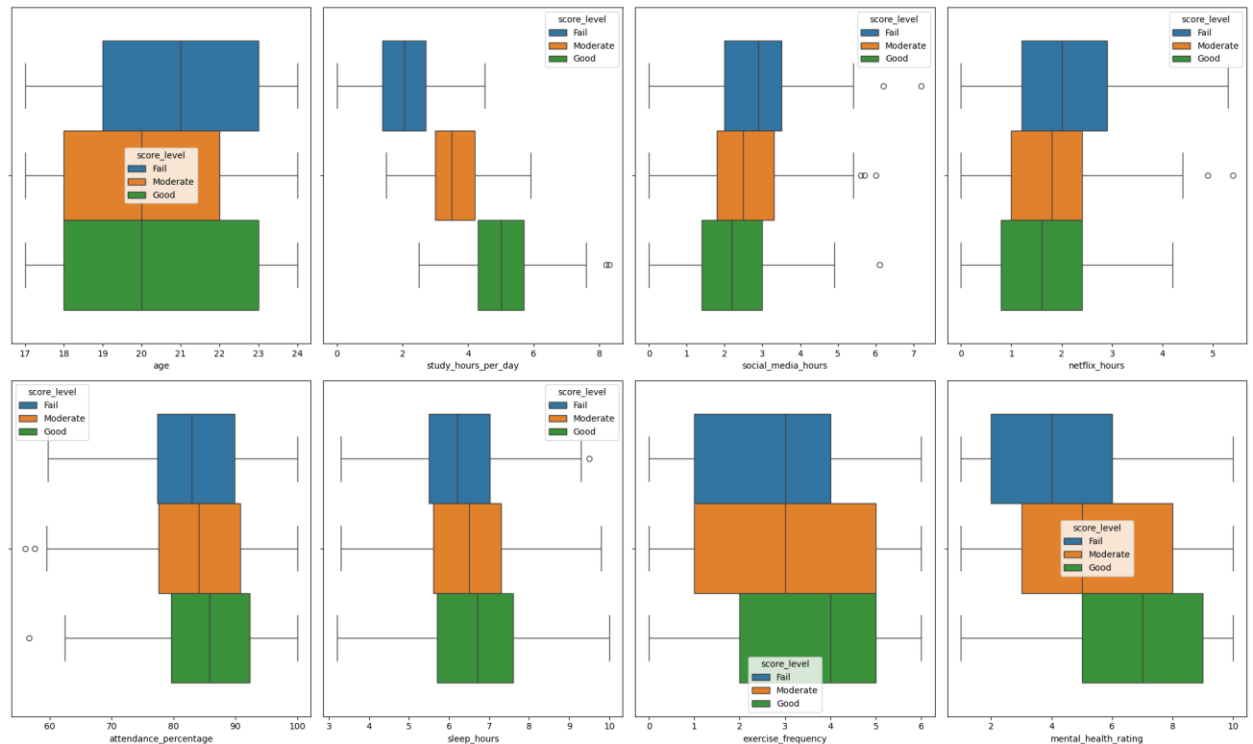


Figure 13 Box plot of Numerical Variables under Different Exam Score Level

Here, the exam scores are separated into 3 levels: `Fail` represents scores under 60; `Moderate` represents scores from 60 to 80; and `Good` represents scores from 80 to 100. This analysis aims to examine the patterns of students in different score level groups. The key question is: Do students with `Good` score levels have significantly different habits compared to students with other score levels?

With the help of box plots (Figure 13), we can clearly observe several important patterns. Students in the `Good` score level group show a lower median for entertainment hours (social media and Netflix hours) and higher medians for other numerical variables (except for age). In other words, they spend less time on entertainment activities like social media and Netflix. In other words, the box plots reveal that high-performing students tend to spend more time studying, have better attendance rates, get more sleep, exercise more frequently, and maintain better mental health ratings.

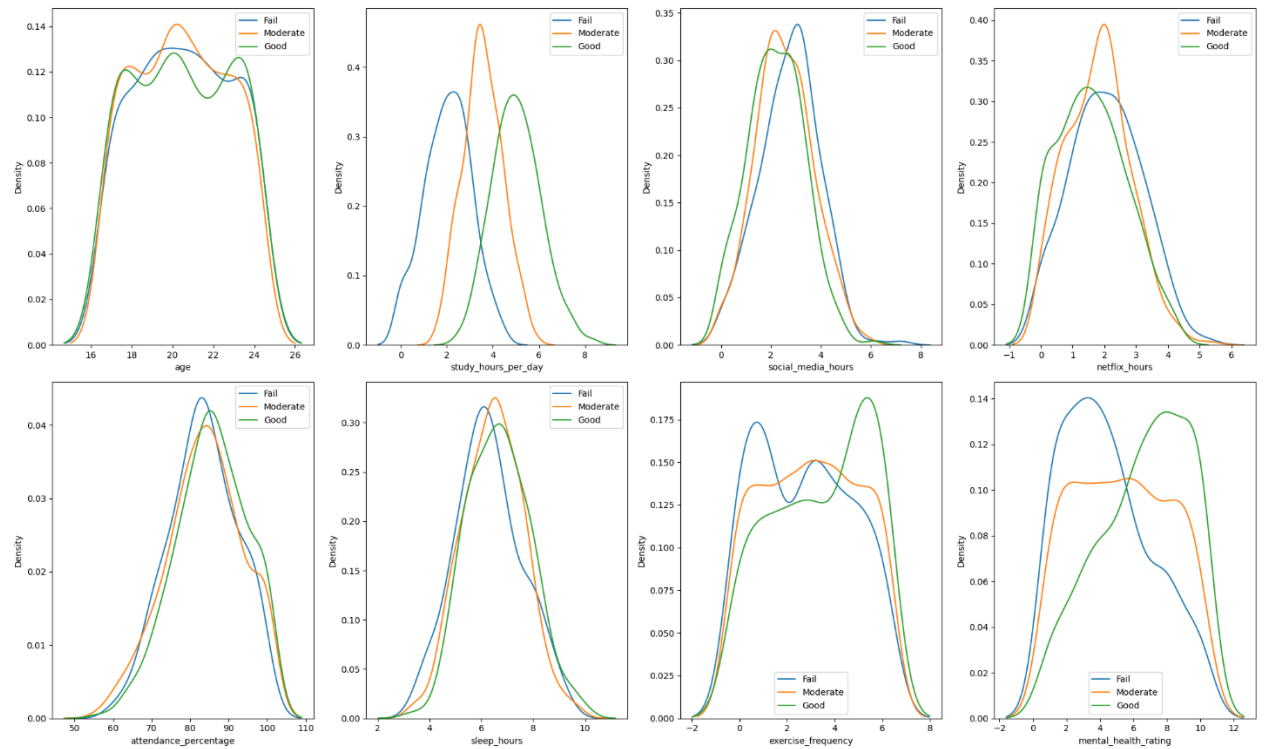


Figure 14 Distribution plot of Numerical Variables under Different Exam Score Level

```

skewness of age under different score levels : -0.00821074139317689
skewness of age under different score levels : 0.021161113267645405
skewness of age under different score levels : 0.0025915910212255615
skewness of study_hours_per_day under different score levels : -0.06946616449290763
skewness of study_hours_per_day under different score levels : 0.1101800398349688
skewness of study_hours_per_day under different score levels : 0.2841919696456917
skewness of social_media_hours under different score levels : 0.011581603499848213
skewness of social_media_hours under different score levels : 0.16198383108047307
skewness of social_media_hours under different score levels : 0.12975144087899684
skewness of netflix_hours under different score levels : 0.12962641634208774
skewness of netflix_hours under different score levels : 0.27218828038431764
skewness of netflix_hours under different score levels : 0.2909388114689299
skewness of attendance_percentage under different score levels : -0.1479533583486621
skewness of attendance_percentage under different score levels : -0.24741673810362963
skewness of attendance_percentage under different score levels : -0.2993210921595659
skewness of sleep_hours under different score levels : 0.1529250305562024
skewness of sleep_hours under different score levels : 0.02498807439446125
skewness of sleep_hours under different score levels : 0.17496567841064053
skewness of exercise_frequency under different score levels : 0.1450035953716374
skewness of exercise_frequency under different score levels : -0.007944892685809527
skewness of exercise_frequency under different score levels : -0.26507378907233864
skewness of mental_health_rating under different score levels : 0.5103627252268884
skewness of mental_health_rating under different score levels : 0.04978127034567049
skewness of mental_health_rating under different score levels : -0.460013704559045

Result: At the  $\alpha=0.05$  level, age has no significant difference under different exam score levels
Result: At the  $\alpha=0.05$  level, study_hours_per_day has significant difference under different exam score levels
Result: At the  $\alpha=0.05$  level, social_media_hours has significant difference under different exam score levels
Result: At the  $\alpha=0.05$  level, netflix_hours has significant difference under different exam score levels
Result: At the  $\alpha=0.05$  level, attendance_percentage has significant difference under different exam score levels
Result: At the  $\alpha=0.05$  level, sleep_hours has significant difference under different exam score levels
Result: At the  $\alpha=0.05$  level, exercise_frequency has significant difference under different exam score levels
Result: At the  $\alpha=0.05$  level, mental_health_rating has significant difference under different exam score levels

```

Figure 15 Skewness and Results of Hypothesis Testing – 2

Based on the box plots and distribution plots (Figure 14), hypothesis tests can be conducted to determine whether there are statistically significant differences between

different score levels for each numerical variable. The methods for conducting these hypothesis tests are the same as for categorical variables.

From the results of the hypothesis tests (Figure 15), we can see that, except for the age attribute, all other numerical variables show significant differences between different score levels. This finding suggests that there are indeed significant differences in the habits of high-performing and low-performing students, where although the overall correlation between exam scores and individual numerical variables may be low, students with `Good` level exam scores still have better study habits overall.

2.5 EDA Conclusion

First, the dataset contains records of 1,000 students and is of good quality overall. Only the `parental_education_level` column has 9.1% missing values and no duplicate records. All numerical variables are relatively normally distributed, but there are some outliers, especially in terms of study hours and entertainment hours.

Second, the strongest positive correlation is found between study time and exam scores, which confirms the basic theory that more time spent studying usually leads to better academic performance. For other variables, although the correlation between a single variable and exam scores may not be high, when multiple habit factors are considered together, their impact on academic performance becomes significant. By dividing students into three levels according to their scores (Fail, Moderate, and Good), we found that all numerical variables, except age, were significantly different across different score levels, where the `Good` student group shows better study habit patterns. These findings provide an empirical basis for developing educational and student support strategies.

CHAPTER 3. Features Engineering

3.1 Handling Missing Values

In the `parental_education_level` column, there are about 9% of the records are missing. Therefore, It is just filled it with the mode. The mode is the category with the most occurrences and filling it in can maintain the original distribution of the entire column.

3.2 Encoding

It is used a small dictionary to map ordinal category variables such as 'Poor-Fair-Good' to integers from 1 to 3, which not only retains the order information but also avoids too many dummy variables. The gender, part-time job, and extracurricular activities columns have no obvious order, so it is used 'get_dummies' to turn them into 0/1 columns.

```
[25] diet_quality = {'Fair': 2, 'Good':3, 'Poor':1}
      parental_education_level = {'Master': 3, 'High School': 1, 'Bachelor': 2}
      internet_quality = {'Average': 2, 'Poor': 1, 'Good': 3}

df_clean['diet_quality'] = df_clean['diet_quality'].map(diet_quality)
df_clean['parental_education_level'] = df_clean['parental_education_level'].map(parental_education_level)
df_clean['internet_quality'] = df_clean['internet_quality'].map(internet_quality)
df_clean = pd.get_dummies(data=df_clean, columns=['gender', 'part_time_job', 'extracurricular_participation'])
df_clean.head()
```

Figure 16 Code of Encoding Category Variables

3.3 New Features

It is added two types of new features. The first type is to add social media hours and Netflix hours to get entertainment hours, because they are essentially "entertainment time", and the combination can more intuitively show how much time students spend on entertainment.

The second type is to divide the exam score into levels (Fail, Moderate, Good), just like what is done in the EDA section, and perform a Z-score on the study time of each student in his/her group of scores, to get 'study_hours_per_day_Dev_score_level'. In this way, it can show who has studied more than the average and who has studied less than the average in the same grade level, which can better reflect the relative degree of effort than simply the total study time.

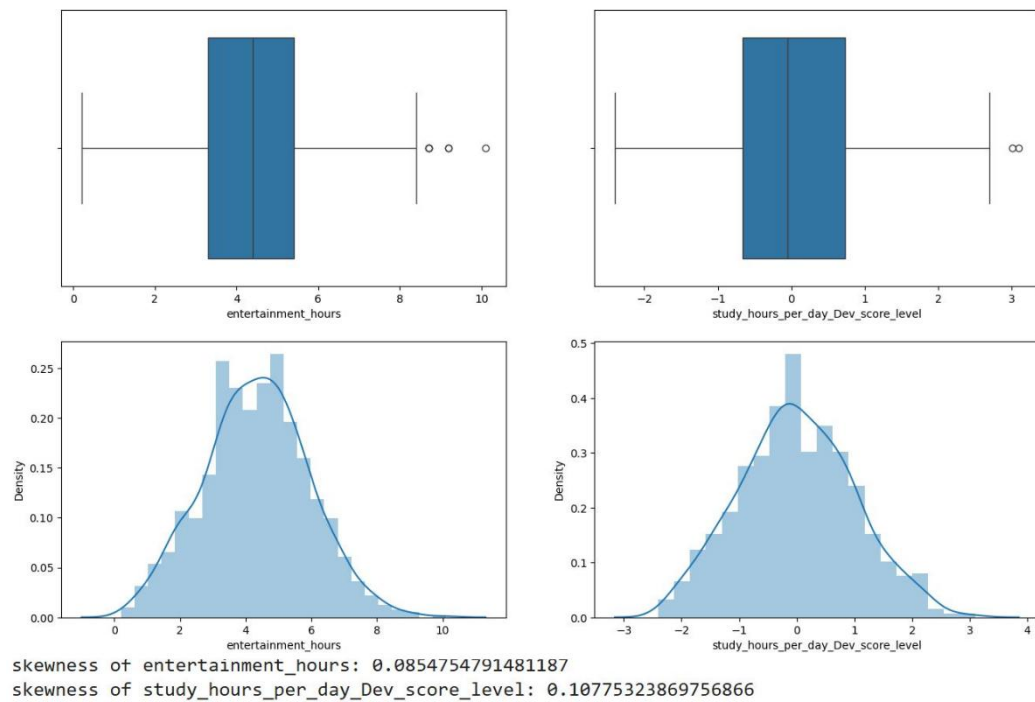


Figure 17 Box plot, Distribution plot and skewness of New Features

	study_hours_per_day	mental_health_rating	exam_score
study_hours_per_day	1.000000	-0.003768	0.825419
social_media_hours	0.020282	0.001496	-0.166733
netflix_hours	-0.031158	0.008034	-0.171779
attendance_percentage	0.026264	-0.018745	0.089836
sleep_hours	-0.027757	-0.006508	0.121683
diet_quality	0.033377	0.027362	0.015018
exercise_frequency	-0.028701	-0.000242	0.160107
parental_education_level	-0.005389	-0.039125	-0.021647
internet_quality	0.014459	-0.048283	-0.036298
mental_health_rating	-0.003768	1.000000	0.321523
exam_score	0.825419	0.321523	1.000000
gender_Female	0.022087	0.005149	0.007977
gender_Male	-0.025634	-0.006279	-0.013191
gender_Other	0.008814	0.002810	0.012976
part_time_job_No	0.029133	-0.013539	0.026608
part_time_job_Yes	-0.029133	0.013539	-0.026608
extracurricular_participation_No	0.003264	0.004741	-0.000881
extracurricular_participation_Yes	-0.003264	-0.004741	0.000881
entertainment_hours	-0.006075	0.006496	-0.237631
study_hours_per_day_Dev_score_level	0.650122	-0.342375	0.197630

Figure 18 Correlation between New Features and Exam Score

```
['study_hours_per_day',  
 'social_media_hours',  
 'netflix_hours',  
 'attendance_percentage',  
 'sleep_hours',  
 'diet_quality',  
 'exercise_frequency',  
 'parental_education_level',  
 'internet_quality',  
 'mental_health_rating',  
 'exam_score',  
 'gender_Female',  
 'gender_Male',  
 'gender_Other',  
 'part_time_job_No',  
 'part_time_job_Yes',  
 'extracurricular_participation_No',  
 'extracurricular_participation_Yes',  
 'entertainment_hours',  
 'study_hours_per_day_Dev_score_level']
```

Figure 19 Final Select Features

CHAPTER 4. References

[1] J. Nath, "Student Habits vs Academic Performance," Kaggle, 2024. [Online]. Available: <https://www.kaggle.com/datasets/jayaantanaath/student-habits-vs-academic-performance>.